Unix, the terminal and Git

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Based on 2024 slides by Chetan Chawla

What is Unix?

Unix is the first portable operating system, that is software that manages your computer's resources for your programs

MacOs is based on Unix, and Linux distributions are Unix-like, meaning it works really similarly to Unix.

What is the Unix Shell?

The shell is the program in which we type commands to communicate with the operating system. BASH (Bourne Again Shell) is the most common Unix Shell

How do we communicate with the Shell?

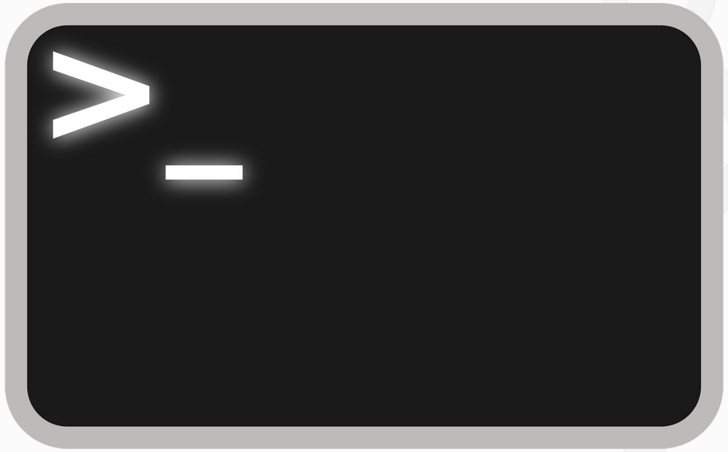
We use a terminal emulator (simply called terminal in most places)

We type commands to perform different actions, like creating and editing files

We will use Jupyter Notebook for most homework. We type can type `%%bash` to make our cell behave like a terminal.

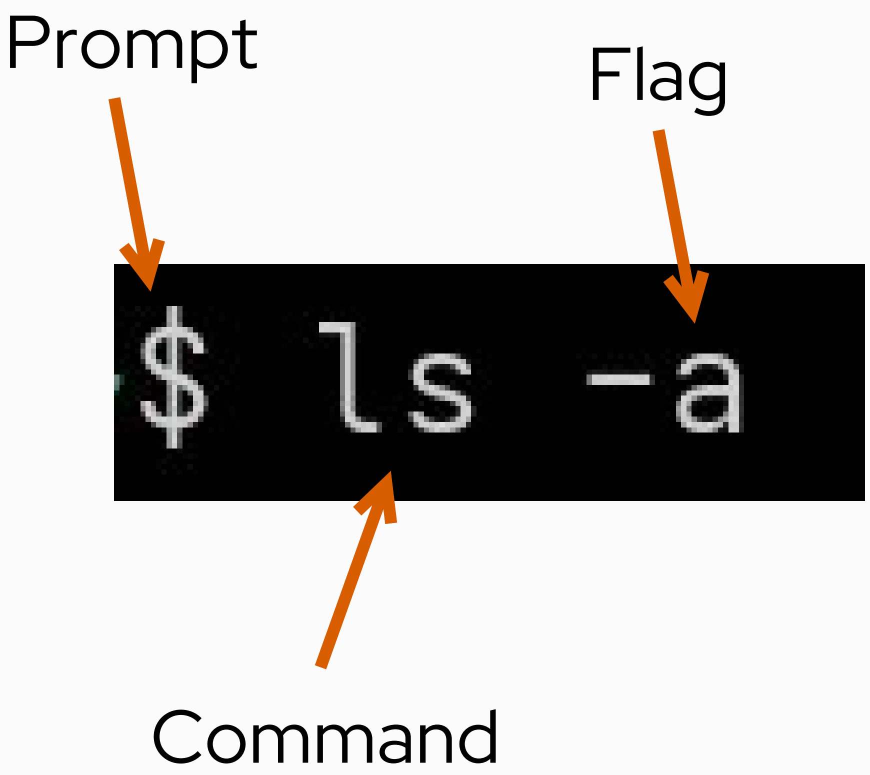
How to access the terminal

- In Linux distributions and MacOs the terminal is pre-installed
- For Windows, you should install WSL (instructions in email you recieved)



Using the terminal

- The terminal uses $ as a prompt for us to type our command
- We can type our command and any "flags" that modify its behavior



Navigating through our files

– **cd location/we_want/to_go**

Cd changes our location. Using cd by itself will always bring us back home

(We can verify our location with the command pwd (print working directory)

– **cd ..**

Takes us to the parent folder (called directories in unix)

– **ls (list)**

Lists all files in a directory

– **ls -a**

Shows us hidden files (starting with a dot)

Creating and editing files

–**mkdir [directoryname]**
Creates a directory at the location we are at

–**touch [filename.txt]**
Touch creates files directly from the terminal

– **nano [filename.txt**]
Edit our file from the terminal

Creating and editing files

– **rm [filename.txt]**

Deletes a specific file or empty directory

The flag -r means recursive and its used to delete directories and their contents

– **cp [filename.txt]**

Copies a file

– **mv [filename.txt] [place/we/moveto]**

Moves a file to a specified place

Creating and editing files

– **nano [filename.txt]**

Edits a file directly on the terminal

– **cat [filename.txt]**

Prints the contents of a file in our terminal

Wildcards

We can use * to mean any character, for example:

– cd Ar* to go into the first directory that starts with the letters Ar
– rm *.txt to remove all .txt files in the directory

Why should I use the terminal?

- It is quicker! (even if it seems slower right now)
- You can execute scripts, automatizing a task you do often
- Some programs can only be run in a terminal

Introduction to Git

Git is a version control system, used for keeping track of changes in files in a project.

This helps us go back to versions we liked better or work on multiple versions of the same files without losing information.

How to make a repository

- **git init**

Initialize a repository in your current directory

- **git add [filename]**

Adds a file (and starts keeping track of its changes)

- **git reset [filename]**

Stops tracking a file

- **git commit –m 'message'**

Commits the changes in all the tracked files (writes down all the changes you made)

Collaborating

We can upload our work in public repositories, where other people can work with us and share our work.

This is done with the command git push.

Assignments

Assignment 1: Terminal Practice

- Create a new directory named foo_dir
- Change directories to foo_dir
- Write "Hello, world" to a text file
- Display the contents of the text file
- Make a copy of the text file
- Place the copy in a new sub-directory, foo_sub_dir

Assignment 2: Complete the Software Carpentry Course for Unix

Assignments

Assignment 3: Create a GitHub account

(If you have one already you don't have to make a new one)

Assignment 4: Git practice

- Initialize a repository in the foo_dir directory you created in part 1
- Add the file where you wrote "Hello, world" and it's copy to the tracked files
- Make a commit of this files, adding the text "My first commit" to the message

Thank you for your attention!